General Chemistry I
Recitation 1

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QQ Group and recitation information

2025秋-普通化学

群号：1036982311

QQ discussion group: 1036982311

Please use your real name

Recitation: 19:50-20:35, Monday
Teaching center 204

学年学期: 2025-2026学年1学期 切换学期

CHEM1103.03[普通化学I]课程安排

节次/周次	星期一	星期二	星期三	星期四	星期五
第一节 8:15 - 9:00					
第二节 9:10 - 9:55					
第三节 10:15 - 11:00			普通化学I(CHEM1103.03) (谢研) (单1-15,教学中心204)		普通化学I(CHEM1103.03) (谢研) (1-16,教学中心204)
第四节 11:10 - 11:55					
第五节 13:00 - 13:45					
第六节 13:55 - 14:40					
第七节 15:00 - 15:45					
第八节 15:55 - 16:40					
第九节 16:50 - 17:35					
第十节 18:00 - 18:45					
第十一节 18:55 - 19:40					
第十二节 19:50 - 20:35	普通化学I(CHEM1103.03) (课程助教) (2-16,教学中心204)				
第十三节 20:45 - 21:30					

What recitations cover

0. “Environment setup”

1. Answers and solutions to PS, quizzes and midterms.

2. Review/supplement of lectures.

3. Additional exercises/knowledge.

4. Relevant/irrelevant stuff…

Something you should know

- Attending recitation is not compulsory, but highly recommended.
- PS should be submitted to BB.
- Key of PS, quizzes will not be posted when the recitation material is on BB.
- Bring a calculator with you.
- Get used to English.

Significant figures(有效数字)

https://en.wikipedia.org/wiki/Significant_figures

The number of *significant figures* is the number of digits used to express a measured or calculated quantity, excluding zeros that may precede the first nonzero digit.
(第一位非零数字起始，直到末尾数字终止)

Non-zero digits within the given measurement or reporting resolution are significant.

Zeros between two significant non-zero digits are significant.

Zeros to the left of the first non-zero digit are not significant.

Zeros to the right of the last non-zero digit in a number with the decimal point are significant if they are within the measurement or reporting resolution.

A mathematical or physical constant has significant figures to its known digits.

E.g. 9.22 0.922 9.22000 0.00000000922 0.000092020 π 6.02×10^{23}

Significant figures(有效数字)

Rounding to significant figures（四舍“五”入）

To round a number to n significant figures, these rules should be followed:

If the $n + 1$ digit is less than 5, keep the n digit.（四舍）

1.344→1.34 1.34→1.3 2.8181812→2.818181

If the $n + 1$ digit is greater than 5, add 1 to the n digit.（六入）

1.346→1.35 1.36→1.4 9.79797979→9.7979798

If the $n + 1$ digit is 5, round half to even, which rounds to the nearest even number.（五成双）

1.235→1.24 1.225→1.22 2.565656565→2.56565656 2.575757575→2.57575758

For an integer in rounding, replace the digits after the n digit with zeros（整数数位保留）

1233→1230→1200

Significant figures(有效数字)

Addition and subtraction of significant figures（加减最左）

The last significant figure position in the calculated result should be the same as the leftmost or largest digit position among the last significant figures of the measured quantities in the calculation.以小数点后位数最少的为准（最左）

$1.234 + 2 = 3.234 \approx 3$ $1.234 + 2.0 = 3.234 \approx 3.2$ $0.01234 + 2 = 2.01234 \approx 2$

Multiplication and division（乘除最短）

The calculated result should have as many significant figures as the least number of significant figures among the measured quantities used in the calculation.

直接取最少位数的有效数字作为结果有效数字

$1.234 \times 2.0 = 2.468 \approx 2.5$ $0.012345678 / 0.00234 = 5.2759 \approx 5.28$

Significant figures(有效数字)

Logarithm and antilogarithm（指对运算，方去对来）

The base-10 logarithm of a normalized number (i.e., a $\times 10^{b}$ with $1 \leq a < 10$ and b as an integer), is rounded such that its decimal part as many significant figures as the significant figures in the normalized number.

对数运算： 结果的小数点后位数和真数有效数字位数相同（多一位）

$\log_{10}(3.000 \times 10^4) = \log_{10}(10^4) + \log_{10}(3.000) = 4.000000...$ (exact number so infinite significant digits) + $0.4771212547... = 4.4771212547 \approx 4.4771.$

pH=2.34

When taking the antilogarithm, the result is rounded to have as many significant figures as the significant figures in the decimal part of the number to be antiloged.

指数运算： 结果的有效数字位数和指数小数点后位数相同（少一位）

$10^{4.4771} = 29998.5318119... = 30000 = 3.000 \times 10^4$

Round only on the final calculation result(as possible as you can), Or the rounding error might be accumulated.

Significant figures(有效数字)

Logarithm and antilogarithm（指对运算）补充：

“方去对来的本质实际上来源于对数运算的性质”

$\log_{10}(3.000 \times 10^4)$ = $\log_{10}(10^4)$ + $\log_{10}(3.000)$ = 4.000000... (exact number so infinite significant digits) + 0.4771212547... = 4.4771212547 ≈ 4.4771.

在本例中需要把3.000×10^4分为3.000 和 10^4两部分来讨论，由于对两个相乘数取对数操作应该是两个底数取对数后再相加，故需要遵循加法的有效数字修约准则。对于10^4取对数后是精确的，没有误差的4（由对数的定义得知），则这里的“4”是带着无限位有效数字的精确值。而本例子的有效数字选择则只由3.000的精确度决定。

所以对数的整数部分（指数的“10的整数次方部分”不影响有效数字）

$$e^{1.14414} * \lg(1.332)/10^{1.0086} = 0.0383246129$$

依照乘法规则，有效数字和最少有效数字位数的“1.332”保持一致，应是4位有效数字

0.03832（“方去对来”）

Solutions of exercises

Calculate the relative atomic mass of carbon, taking the relative atomic mass of ^{13}C to be 13.003354 on the ^{12}C scale.

Solution

Set up the following table:

Isotope	Isotopic Mass × Abundance
^{12}C	$12.000000 \times 0.98892 = 11.867$
^{13}C	$13.003354 \times 0.01108 = 00.144$

Chemical relative atomic mass = 12.011

Solutions of exercises

Radon-222 (^{222}Rn) has recently received publicity because its presence in basements may increase the number of cancer cases in the general population, especially among smokers. State the number of electrons, protons, and neutrons that make up an atom of ^{222}Rn.

Solution

From the table on the inside front cover of the book, the atomic number of radon is 86; thus, the nucleus contains 86 protons and $222 - 86 = 136$ neutrons. The atom has 86 electrons to balance the positive charge on the nucleus.

Solutions of exercises

One of the heaviest atoms found in nature is ^{238}U. Its relative atomic mass is 238.0508 on a scale in which 12 is the atomic mass of ^{12}C. Calculate the mass (in grams) of one ^{238}U atom.

Solution

Because the mass of N_A atoms of ^{238}U is 238.0508 g and N_A is 6.0221420×10^{23}, the mass of one ^{238}U atom must be

$$\frac{238.0508\ \text{g}}{6.0221420 \times 10^{23}} = 3.952926 \times 10^{-22}\ \text{g}$$

Solutions of exercises

Nitrogen dioxide (NO_2) is a major component of urban air pollution. For a sample containing 4.000 g NO_2, calculate (a) the number of moles of NO_2 and (b) the number of molecules of NO_2.

Solution

(a) From the tabulated molar masses of nitrogen (14.007 g mol^{-1}) and oxygen (15.999 g mol^{-1}), the molar mass of NO_2 is

$$14.007 \text{ g mol}^{-1} + (2 \times 15.999 \text{ g mol}^{-1}) = 46.005 \text{ g mol}^{-1}$$

The number of moles of NO_2 is then

$$\text{mol of NO}_2 = \frac{4.000 \text{ g NO}_2}{46.005 \text{ g mol}^{-1}} = 0.08695 \text{ mol NO}_2$$

(b) To convert from moles to number of molecules, multiply by Avogadro's number:

$$\text{molecules of NO}_2 = \left(0.08695 \text{ mol NO}_2\right) \times 6.0221 \times 10^{23} \text{ mol}^{-1}$$
$$= 5.236 \times 10^{22} \text{ molecules NO}_2$$